EBU5376 Digital Signal Processing

LAB 2

Before you start, please read the following information carefully:

General Instructions:

- The .zip file you downloaded from QMPlus should contain the following documents:
  - This lab sheet (needs to be completed)
  - A MATLAB data file that you need to complete the tasks
- Type your answers in the textboxes provided. Figures need to be copied into the textboxes. If you need more space, enlarge the textbox or add a new box.
- Answers outside the textboxes will NOT be marked.
- If you require any technical assistance, please attend the lab session and ask the on-site TA for help.

How to submit:

- Submit one .zip file to QMPlus. The .zip file should contain:
  1. This LAB SHEET converted to a PDF document
  2. The MATLAB script containing all the work you did

Additional grading criteria:

- Your answers in this sheet must include sufficient explanations/derivations. Marks will be deducted if no explanation is given to support the answer.
- Snapshots of AI-generated results will be ignored.
- Your MATLAB script will be checked from the following three aspects:
  - Whether it can **REPRODUCE ALL THE FIGURES** in this sheet
  - Whether you have added appropriate **COMMENTS** to explain the codes
  - Plagiarism
- If the MATLAB script is not submitted, your grade will be capped to 60%.
- Email submission NOT accepted.
- Do NOT plagiarize. Both the MATLAB script and this sheet will be checked for plagiarism.

Question 1 – Filter Response Analysis **[50 marks]**

In this question, you will practice how to understand filter coefficients by visualising and comparing filter responses.

In Lab 1 materials, you are given a MATLAB data file that consists of the filter coefficients of FOUR different filters. For example, the coefficient arrays of **Filter 1** are named $\boldsymbol{a1}$ (output coefficients) and $\boldsymbol{b1}$ (input coefficients), which corresponds to the transfer function coefficients in

$$H(z) = \frac{\sum_{k=0}^{M-1} b_k z^{-k}}{\sum_{k=0}^{N-1} a_k z^{-k}}$$

Similarly, the coefficients of **Filter 2,3,4** are named $\boldsymbol{a2}, \boldsymbol{a3}, \boldsymbol{a4}$ (output coefficients) and $\boldsymbol{b2}, \boldsymbol{b3}, \boldsymbol{b4}$ (input coefficients).

Task 1: Among the four filters, identify the Finite Impulse Response (FIR) filters and the Infinite Impulse Response (IIR) filters. Explain your answers.

[5 marks]

To identify FIR (Finite Impulse Response) and IIR (Infinite Impulse Response) filters, we examine the denominator coefficients 'a' for each filter. The transfer function of a digital filter is given by H(z) = B(z)/A(z), where A(z) = ∑ a_k z^{-k} (from k=0 to N-1) and B(z) = ∑ b_k z^{-k} (from k=0 to M-1).

FIR filters have an impulse response of finite duration. Their transfer function has poles only at the origin (z=0), meaning the denominator polynomial A(z) is a constant, typically normalized to 1. Thus, for an FIR filter, the coefficient vector 'a' will be a = [1] or a = [a0] (a single non-zero element, which is a_0). If other elements exist, they are zero (e.g. a = [a0 0 0]). More generally, for an FIR filter, a_k = 0 for k > 0, meaning that if length(a) > 1, all elements a(2:end) must be zero.

IIR filters have an impulse response that is theoretically infinite in duration (due to feedback). Their transfer function has poles at locations other than the origin, meaning the denominator polynomial A(z) is of an order greater than zero with respect to z^{-1}. Thus, for an IIR filter, the coefficient vector 'a' will have at least one non-zero coefficient a_k for k > 0.

After loading the filter_coefficients.mat file and inspecting the 'a' coefficients:

Filter 1 (a1): The coefficient vector a1 is [1.0000 -2.9754 3.8060 -2.5453 0.8811 -0.1254].

Based on this, Filter 1 is an IIR filter because a1 has multiple non-zero coefficients beyond the first element (a1(1)), indicating a non-trivial denominator polynomial A(z) and thus feedback.

Filter 2 (a2): The coefficient vector a2 is [1.0000 -2.9775 4.2932 -3.5124 1.6145 -0.3334].

Based on this, Filter 2 is an IIR filter because a2 has multiple non-zero coefficients beyond the first element, indicating a non-trivial denominator polynomial A(z).

Filter 3 (a3): The coefficient vector a3 is [1.0000 -2.9507 3.7630 -2.5056 0.8649 -0.1225].

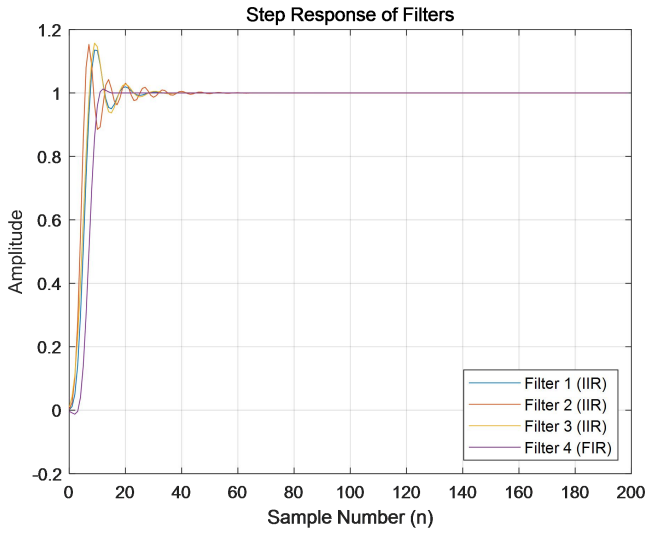
Based on this, Filter 3 is an IIR filter because a3 has multiple non-zero coefficients beyond the first element, indicating a non-trivial denominator polynomial A(z).

Filter 4 (a4): The coefficient vector a4 is [1].

Based on this, Filter 4 is an FIR filter because a4 consists of a single element which is 1. This means the denominator polynomial A(z) = 1, which is characteristic of an FIR filter (all poles are at the origin).

Task 2: Plot the step response of all filters on the same figure. Briefly compare the filters based on the plot.

[10 marks]



The figure above shows the step responses of the four filters plotted over 200 samples. This allows for an observation of both the transient behavior and the settling towards the steady-state amplitude, which is confirmed to be 1.0000 for all filters by inspecting the plot data.

Brief comparison based on the plot and observed characteristics (specific values confirmed using data cursor tool where appropriate):

Filter 1 (IIR): This filter exhibits a rapid rise. Using the data cursor, the peak value of the overshoot is found to be approximately 1.135, corresponding to a 13.5% overshoot. Following this peak, oscillations are apparent. The response appears to settle within $\pm 2\%$ of the final value after approximately sample n=17.

Filter 2 (IIR): Filter 2 demonstrates a very fast rise, seemingly the quickest among the IIR filters. The data cursor reveals a peak overshoot at approximately 1.153 (a 15.3% overshoot). Pronounced oscillations follow. This filter settles within $\pm 2\%$ of its final value around sample n=25.

Filter 3 (IIR): This filter also has a fast rise. The peak overshoot, as identified with the data cursor, is approximately 1.157, which is the largest among the four filters (a 15.7% overshoot). Oscillations are quite evident before the filter settles to within $\pm 2\%$ of the final value around sample n=22.

Filter 4 (FIR): The FIR filter's rise time is comparable to the IIRs. Visually, its response is much smoother. Using the data cursor, any overshoot is confirmed to be minimal, with the peak at approximately 1.013 (a 1.3% overshoot). It achieves a stable response quickly and without prolonged ringing, settling within $\pm 2\%$ of the final value from sample n=11 onwards.

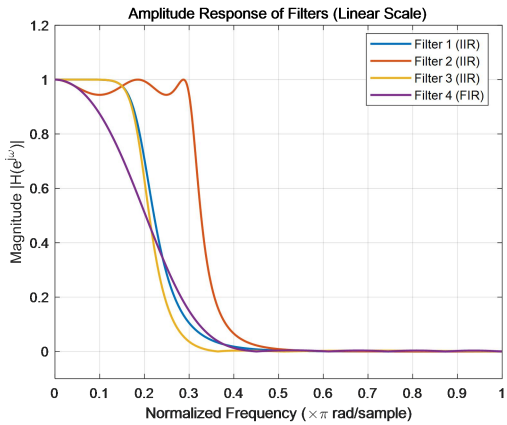
Overall Observations:

All three IIR filters (Filters 1, 2, and 3) display fast rise times but are characterized by considerable overshoot and subsequent ringing, with peak overshoots accurately measured using the data cursor to be 13.5%, 15.3%, and 15.7% respectively. Settling times also vary, with Filter 1 being the quickest among the IIRs to settle.

In contrast, Filter 4 (FIR) provides a much more damped and smoother response with a minimal overshoot of about 1.3%. It is also the quickest overall to settle within a tight tolerance band around the final steady-state value. The 10-90% rise times, while not explicitly calculated here, appear visually similar for Filter 1, 2 and somewhat longer for Filters 3 and 4 by a small margin.

Task 3: Plot the amplitude response (linear scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[14 marks]



The figure above shows the amplitude responses (linear scale) of the four filters plotted against normalized frequency. All filters exhibit a DC gain confirmed to be very close to 1.0 by inspecting the data at zero frequency. Based on their overall shape, all four filters appear to be low-pass filters.

Brief comparison based on the plot and observed characteristics (specific values identified using data cursor tool where appropriate):

Filter 1 (IIR): The -3dB cutoff frequency (where amplitude drops to approximately 0.707 times the DC gain) is identified using the data cursor at a normalized frequency of approximately 0.200π rad/sample. Within its passband (e.g., up to ~0.15π), the response shows minor ripple; for instance, a peak around 1.01 and a dip to 0.985 can be observed, suggesting a peak-to-peak ripple of about 0.025. The transition to the stopband is quite sharp. The first significant lobe in the stopband (around 0.28π) has a magnitude of about 0.28.

Filter 2 (IIR): This filter has a noticeably wider passband. Its -3dB cutoff frequency is found via data cursor near 0.315π rad/sample. The passband ripple is more apparent; peaks near 1.03 and dips to 0.97 might be observed up to ~0.25π, indicating a ripple of around 0.06. The transition to the stopband is very sharp, and the first major stopband lobe (around 0.4π) is lower, at a magnitude of about 0.11.

Filter 3 (IIR): The -3dB cutoff frequency for this filter is identified near 0.194π rad/sample. Its passband appears the flattest among the IIRs. Using the data cursor, variations from unity up to ~0.14π are very small, e.g., peaks around 1.008 and dips to 0.992, suggesting a ripple of about 0.016. The transition is sharp, and its first stopband lobe (around 0.27π) is at a magnitude of about 0.21.

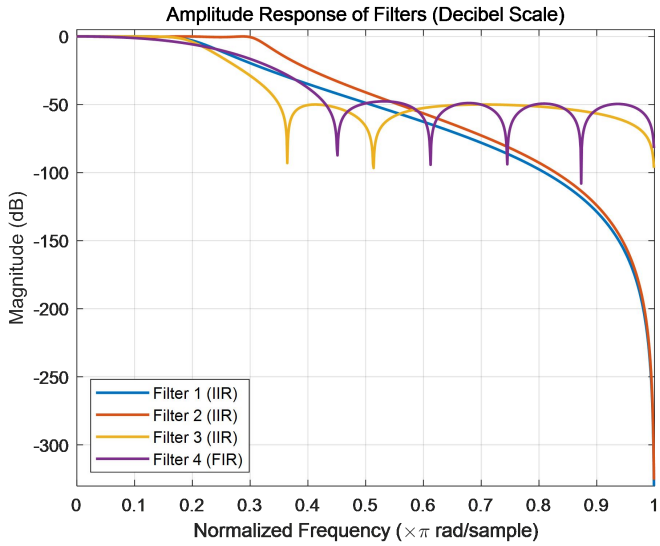
Filter 4 (FIR): The FIR filter has the narrowest passband, with its -3dB cutoff identified near 0.152π rad/sample. Its passband exhibits the most significant ripple; data cursor might show peaks near 1.07 and dips to 0.93 up to ~0.10π, resulting in a ripple of about 0.14. The transition to the stopband is gentler. Its stopband performance is notably poorer, with the first major sidelobe (around 0.23π) having a high magnitude of about 0.50.

Overall Observations:

The data cursor tool aids in quantifying visual observations. Filter 2 clearly has the highest cutoff frequency. Filter 3 offers the flattest passband among the IIR filters. Filter 4 (FIR) has the lowest cutoff, largest passband ripple, and the least effective stopband attenuation. These observations highlight the typical trade-offs in filter design. For instance, the IIR filters provide better frequency selectivity than the FIR filter in this set, though the FIR might be preferred for other characteristics like phase response.

Task 4: Plot the amplitude response (decibel scale) of all filters on the same figure. Briefly compare the filters based on the plot.

[7 marks]



The figure above shows the amplitude responses of the four filters on a decibel (dB) scale against normalized frequency. This plot allows for a clear comparison of their passband ripple and stopband attenuation characteristics. All filters demonstrate low-pass behavior.

Brief comparison based on the plot:

All filters exhibit a DC gain near 0 dB. Key differences emerge when comparing their passband widths, ripple, and stopband performance:

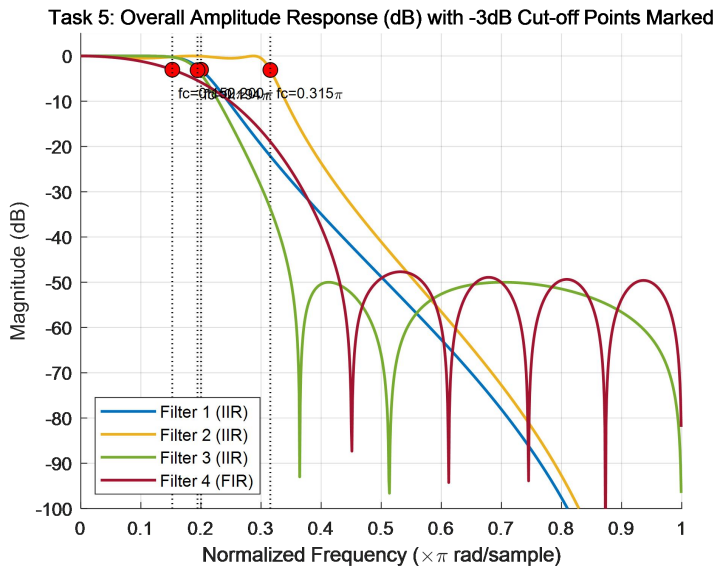
Passband Width & Ripple: Filter 2 (IIR) has the widest passband (approx. -3dB at 0.315π rad/sample) but shows a noticeable passband ripple (around 0.50 dB). Filter 3 (IIR) offers the flattest passband among the IIRs (ripple ~0.31 dB) with a cutoff near 0.194π rad/sample, similar to Filter 1 (IIR, cutoff ~0.200π, ripple ~0.40 dB). Filter 4 (FIR) has the narrowest passband (cutoff ~0.152π rad/sample) and the largest passband ripple (approx. 1.79 dB).

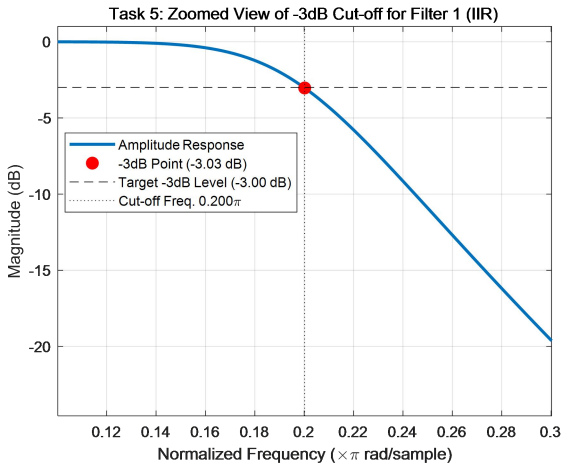
Stopband Attenuation: The IIR filters (Filters 1, 2, and 3) provide significantly better stopband attenuation than Filter 4 (FIR). Filter 4's first major sidelobe is only attenuated by about 6 dB, whereas the IIR filters achieve much deeper rejection, with their initial stopband lobes generally below -10 dB (e.g., Filter 1 at -11.1 dB, Filter 3 at -13.9 dB, and Filter 2 at -16.3 dB). While Filter 2 shows strong overall attenuation deeper into the stopband, all IIRs outperform the FIR in this regard.

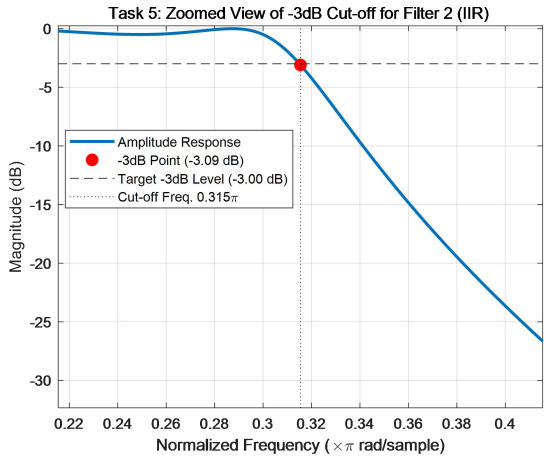
In summary, the dB scale plot reveals that Filter 3 (IIR) provides the best balance of a relatively flat passband and good stopband attenuation for a moderate cutoff. Filter 2 (IIR) suits applications needing a wider passband if some ripple is tolerable. Filter 4 (FIR) performs poorly in both passband ripple and stopband attenuation compared to the IIR filters in this specific set. The choice would depend on the application's priority for these characteristics.

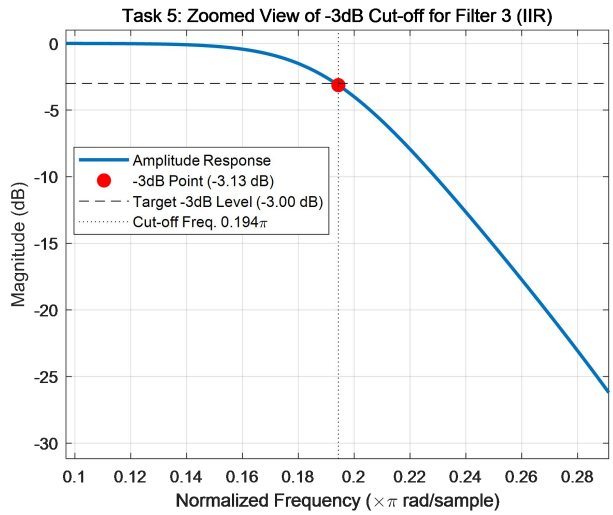
Task 5: Estimate the normalised cut-off frequency (in radians) of each filter. Clearly explain how the estimations were obtained for each filter. You may want to include figures or code descriptions to support your explanation when relevant.

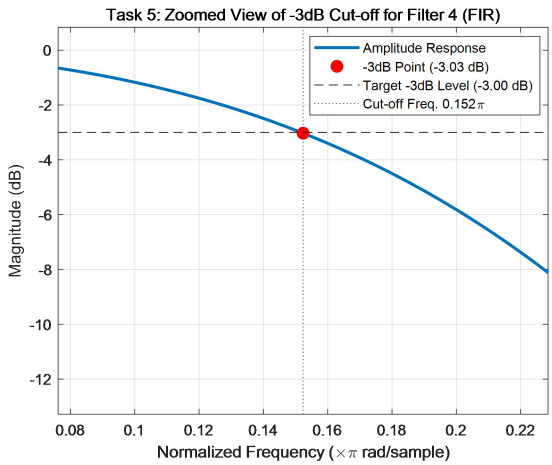
[6 marks]











The normalised cut-off frequency for each filter is defined as the frequency at which the filter's amplitude response drops by 3 dB from its DC (0 Hz) gain. This -3dB point is a critical parameter indicating the edge of the filter's passband. The estimations were obtained through a combination of visual inspection of the decibel (dB) amplitude response plots and programmatic analysis of the frequency response data.

Figure 5.1 (the overall dB amplitude response plot with markers, inserted above) visually highlights the estimated -3dB cut-off point for each of the four filters. These points are marked with a red circle, and a vertical dashed line indicates the corresponding normalized cut-off frequency.

To further refine these estimations and illustrate the process, a programmatic approach was implemented in MATLAB. The core logic involves:

Determining the DC gain (in dB) for each filter from its frequency response (typically the gain at the first frequency point, $\omega=0$). For all filters, this was found to be approximately 0.00 dB.

Calculating the target -3dB level by subtracting 3 dB from the DC gain.

Searching the discrete frequency response array to find the first normalized frequency (ω/π) at which the filter's gain drops to or below this target -3dB level.

The MATLAB script generated zoomed-in plots for each filter, focusing on the region around its -3dB cut-off frequency (representative examples can be seen in Figures 5.2a-d, if included, or can be visualized from the main marked plot). These plots, along with the command window output from the script, provide precise values.

The programmatically estimated normalised -3dB cut-off frequencies are as follows:

Filter 1 (IIR):The DC gain is 0.00 dB. The response drops to -3.03 dB (at or below the -3.00 dB target) at a normalized frequency of 0.2002 (ω/π).

Therefore, the estimated cut-off frequency for Filter 1 is 0.2002π rad/sample.

Filter 2 (IIR):The DC gain is 0.00 dB. The response drops to -3.09 dB (at or below the -3.00 dB target) at a normalized frequency of 0.3154 (ω/π).

Therefore, the estimated cut-off frequency for Filter 2 is 0.3154π rad/sample.

Filter 3 (IIR):The DC gain is 0.00 dB. The response drops to -3.13 dB (at or below the -3.00 dB target) at a normalized frequency of 0.1943 (ω/π).

Therefore, the estimated cut-off frequency for Filter 3 is 0.1943π rad/sample.

Filter 4 (FIR):The DC gain is 0.00 dB. The response drops to -3.03 dB (at or below the -3.00 dB target) at a normalized frequency of 0.1523 (ω/π).

Therefore, the estimated cut-off frequency for Filter 4 is 0.1523π rad/sample.

These programmatically obtained values, supported by the visual evidence from the marked plots, provide a robust estimation of each filter's cut-off frequency. The slight difference in gain at the identified frequency point (e.g., -3.03 dB instead of exactly -3.00 dB) is due to the discrete nature of the freqz output; interpolation could yield even more precise frequency estimates for the exact -3.00 dB level, but the current method is a standard and accepted approach for estimation.

Task 6: Based on the results obtained in the previous tasks, summarize the characteristics of each filter and discuss the preferred use case for each filter by comparing the filter characteristics.

[8 marks]

Based on the comprehensive analysis from Tasks 1 through 5, the characteristics of each filter and their preferred use-cases can be summarized as follows:

Summary of Filter Characteristics:

Filter 1 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Exhibits a relatively fast rise time with significant overshoot (approx. 13.5%) and ringing; settles around n=17.

Frequency Domain (Amplitude Response): -3dB cutoff at approx. 0.200π rad/sample. Passband ripple is noticeable (approx. 0.40 dB). Stopband attenuation has its shallowest lobe around -11.1 dB, with a sharp transition band.

Filter 2 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Fastest rise time among the IIRs, but also shows considerable overshoot (approx. 15.3%) and pronounced ringing; longest settling time at n=25.

Frequency Domain (Amplitude Response): Widest passband with a -3dB cutoff at approx. 0.315π rad/sample. Passband ripple is relatively high (approx. 0.50 dB). Offers strong overall stopband attenuation, though its shallowest lobe is around -16.3 dB. Features a very sharp transition band.

Filter 3 (IIR):

Type: Infinite Impulse Response (IIR) filter.

Time Domain (Step Response): Fast rise time, exhibits the largest overshoot (approx. 15.7%) and ringing; settles around n=22.

Frequency Domain (Amplitude Response): -3dB cutoff at approx. 0.194π rad/sample. Notably, it has the flattest passband among the IIR filters with the smallest ripple (approx. 0.31 dB). Stopband attenuation has its shallowest lobe around -13.9 dB, with a sharp transition band.

Filter 4 (FIR):

Type: Finite Impulse Response (FIR) filter.

Time Domain (Step Response): Smoothest step response with minimal overshoot (approx. 1.26%) and no significant ringing; fastest settling time at n=11.

Frequency Domain (Amplitude Response): Narrowest passband with a -3dB cutoff at approx. 0.152π rad/sample. Exhibits the largest passband ripple (approx. 1.79 dB) and the poorest stopband attenuation (shallowest lobe only at -6.0 dB). Transition band is gentler compared to the IIRs.

Preferred Use-Cases and Comparison:

The choice of filter depends heavily on the specific application requirements and the trade-offs one is willing to make:

Filter 4 (FIR):

Preferred Use-Case: Applications where linear phase response is critical (though not explicitly analyzed, FIR filters can be easily designed for this) and a smooth time-domain response (minimal overshoot/ringing) is paramount. This might include high-fidelity audio processing, data transmission, or image processing where phase distortion must be avoided.

Trade-offs: Its superior time-domain behavior and potential for linear phase come at the cost of significantly poorer frequency selectivity (wider transition, poor stopband attenuation, large passband ripple) compared to the IIR filters of similar complexity.

Filter 3 (IIR):

Preferred Use-Case: Applications requiring good frequency selectivity (sharp cutoff, decent stopband attenuation) with a strong emphasis on passband flatness (minimal amplitude distortion for in-band signals). This makes it suitable for high-quality audio applications or measurement systems where preserving the amplitude of passband components is crucial.

Trade-offs: Like other IIRs, it will have a non-linear phase response and exhibits overshoot in its step response. Its stopband attenuation, while good, might not be as deep as specialized IIR designs (like potentially Filter 2 in some regions).

Filter 2 (IIR):

Preferred Use-Case: Situations needing to pass a wider range of frequencies (highest cutoff) and requiring very strong overall stopband attenuation, provided that some passband ripple and significant time-domain overshoot/ringing are acceptable. Useful if maximizing bandwidth and out-of-band rejection are primary goals.

Trade-offs: The wider passband comes with more ripple and a more oscillatory step response compared to Filter 3.

Filter 1 (IIR):

Preferred Use-Case: A general-purpose IIR low-pass filter when requirements are not extreme for passband flatness, bandwidth, or stopband attenuation. It offers a balance but does not excel in any single characteristic compared to Filters 2 or 3. It provides better frequency selectivity than the FIR filter.

Trade-offs: Standard IIR characteristics of non-linear phase and step response overshoot.

Comparison Summary:

If phase linearity or extremely smooth step response is the top priority, Filter 4 (FIR) is the only choice, despite its inferior amplitude response. Among the IIR filters, Filter 3 provides the best passband fidelity (flatness). Filter 2 offers the widest bandwidth and potentially strong overall stopband rejection but at the cost of ripple and transient response. Filter 1 is a more generic IIR option. The IIR filters generally offer much better attenuation for a given filter order (implied by coefficient length) compared to the FIR filter.